RAG-Powered Biomedical Q&A with Semantic Hallucination Detection:
Comparing OpenAI and PubMedBERT Embedding Models

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ABSTRACT

Large language models (LLMs) are increasingly deployed in clinical decision support, yet they routinely generate plausible-sounding but factually incorrect medical information — a phenomenon known as hallucination. This paper presents a Retrieval-Augmented Generation (RAG) pipeline for biomedical question answering on the PubMedQA dataset, with a novel evaluation framework that uses embedding-based cosine similarity as an automated, scalable hallucination proxy. We systematically compare six experimental conditions across two dimensions: RAG on/off crossed with prompting strategy (Zero-shot vs. Chain-of-Thought), and two embedding models for both retrieval and hallucination scoring — the general-purpose OpenAI text-embedding-3-small and the biomedical domain-specific NeuML/pubmedbert-base-embeddings. Results show that RAG reduces hallucination rates by up to 3.8 times compared to ungrounded baselines, and that domain-specific embeddings produce marginally superior retrieval quality and grounding scores. Critically, the two embedding models disagree most on Chain-of-Thought answers without retrieval, where PubMedBERT rates 86% of answers as hallucinated versus OpenAI's 38% — revealing that general-purpose embedders underestimate clinical language drift in free reasoning chains.

CCS Concepts: Computing methodologies -> Natural language processing; Applied computing -> Health informatics.

Keywords: Retrieval-Augmented Generation, hallucination detection, biomedical NLP, large language models, PubMedQA, clinical AI.

1. INTRODUCTION

Artificial intelligence systems based on large language models have demonstrated remarkable capabilities in clinical question answering, literature synthesis, and diagnostic reasoning. Yet their widespread adoption in healthcare settings is impeded by a well-documented failure mode: hallucination. LLMs generate medically plausible but factually incorrect statements with confidence, presenting a direct patient safety risk. A model that fabricates a drug interaction, misquotes a diagnostic threshold, or invents a clinical guideline does not merely provide suboptimal advice — it provides actively dangerous advice to clinicians who may lack the contextual knowledge to detect the error.

The clinical stakes are high. Medical errors are estimated to be the third leading cause of death in the United States (Makary and Daniel, 2016), and AI-generated misinformation that enters clinical workflows could accelerate this toll. Despite this urgency, most existing benchmarks for biomedical LLMs evaluate only final-answer accuracy — a binary measure that fails to capture whether correct answers were reached through faithful reasoning grounded in evidence, or through confident confabulation that happened to arrive at the right conclusion.

This paper addresses two interconnected problems. First, can Retrieval-Augmented Generation (RAG) meaningfully reduce hallucination in biomedical question answering compared to ungrounded GPT-

4 inference? Second, does the choice of embedding model — general-purpose versus biomedical domain-specific — affect both the quality of evidence retrieval and the sensitivity of automated hallucination detection?

We construct an end-to-end experimental pipeline over the PubMedQA dataset using GPT-4o for answer generation, FAISS for semantic retrieval, and two embedding models for both indexing and hallucination scoring: OpenAI text-embedding-3-small and NeuML/pubmedbert-base-embeddings. We evaluate across six conditions and five metrics, producing a comprehensive characterization of the accuracy-faithfulness trade-off across prompting strategies and embedding models.

Our contributions are: (1) a novel embedding-based hallucination scoring metric that operates without human annotation; (2) a systematic comparison of domain-specific versus general-purpose embeddings for both retrieval and hallucination detection; and (3) the finding that Chain-of-Thought reasoning without retrieved evidence produces clinically imprecise language that general-purpose embedders fail to detect as hallucinated, while biomedical-specific embedders flag it more than twice the rate.

2. RELATED WORK

2.1 Retrieval-Augmented Generation

Lewis et al. (2020) introduced Retrieval-Augmented Generation as a framework that combines parametric memory (stored in neural network weights) with non-parametric memory (retrieved from an external knowledge store) at inference time. In the RAG framework, a retriever encodes both a query and a corpus of documents into a shared embedding space, retrieves the most relevant passages using nearest-neighbor search, and passes them as context to a generator that conditions its output on the retrieved evidence. Lewis et al. demonstrated that RAG outperformed pure parametric models on knowledge-intensive NLP tasks by reducing reliance on memorized but potentially stale or incorrect facts. Our work directly extends this framework to the clinical domain, using FAISS as the retrieval backend and PubMedQA abstracts as the knowledge corpus.

2.2 Clinical LLM Benchmarking

Singhal et al. (2023) presented Med-PaLM 2, a large language model fine-tuned on medical data and evaluated on the MedQA benchmark, achieving performance comparable to human physicians on US Medical Licensing Examination-style questions. Critically, their work highlighted that raw accuracy on multiple-choice benchmarks is insufficient for clinical deployment evaluation — they introduced physician panel evaluation of long-form answers on dimensions including accuracy, reasoning, and potential for harm. This insight directly motivates our LLM-as-Judge evaluation framework, which scores answers on grounding, reasoning quality, and safety rather than accuracy alone. Singhal et al. did not, however, measure hallucination rates systematically, which our work addresses through embedding-based similarity scoring.

2.3 RAG Evaluation Frameworks

Es et al. (2023) proposed RAGAS (Retrieval Augmented Generation Assessment), an automated evaluation framework for RAG systems that measures faithfulness (whether claims in the answer are supported by the retrieved context), answer relevance (whether the answer addresses the question), and context precision (whether retrieved passages are relevant). RAGAS uses an LLM as a judge to decompose answers into atomic claims and verify each against the retrieved context. Our hallucination scoring approach is complementary to RAGAS: rather than decomposing answers into claims, we use embedding space proximity as a continuous, model-based faithfulness signal. This approach is computationally cheaper and does not require a secondary LLM call for each answer, making it more scalable. Comparing RAGAS scores to embedding-based scores on the same dataset is identified as a direction for future work.

3. METHODOLOGY

3.1 Dataset

We use PubMedQA (Jin et al., 2019), a biomedical question-answering dataset consisting of 1,000 expert-labelled question-answer pairs drawn from PubMed abstracts. Each record contains a research question, the source abstract as context, a detailed long-form expert answer, and a ternary label (yes/no/maybe). We restrict evaluation to yes/no questions only (890 records), as the binary classification task provides unambiguous ground-truth labels for accuracy evaluation. From this pool, we sample a stratified evaluation set of 100 questions (50 yes/50 no) using a fixed random seed for reproducibility. The full 890-passage corpus is indexed for retrieval, ensuring that the evaluation questions must compete with 890 candidate passages rather than being retrieved from a closed set — a design choice that produces realistic retrieval difficulty.

3.2 Embedding Models

Two embedding models are compared throughout all pipeline stages:

- OpenAI text-embedding-3-small: A general-purpose dense embedding model producing 1,536-dimensional vectors. Accessed via the OpenAI API. Trained on large-scale web text with no domain-specific fine-tuning for biomedicine.
- NeuML/pubmedbert-base-embeddings: A sentence-level embedding model based on Microsoft BiomedBERT, fine-tuned for semantic similarity on PubMed abstracts using sentence-transformers. Produces 768-dimensional vectors. Runs locally with no API cost, making it readily deployable in resource-constrained settings.

Both models are used for two separate purposes: (1) building and querying the FAISS retrieval index, and (2) scoring hallucination by measuring the cosine similarity between generated answers and source passages.

3.3 Pipeline Architecture

Figure 1 illustrates the full experimental pipeline. The pipeline proceeds in four stages:

- Knowledge Base Construction: All 890 PubMedQA abstracts are embedded using each model and stored in separate FAISS IndexFlatIP indexes, which perform exact inner-product search equivalent to cosine similarity on L2-normalised vectors.
- Retrieval: For each evaluation question, a query embedding is computed and the top-K=3 most similar passages are retrieved from the corresponding FAISS index.
- Answer Generation: GPT-4o generates answers under six experimental conditions. Retrieval-augmented prompts include the top-3 passages as evidence; ungrounded prompts do not. Chain-of-Thought prompts direct the model to reason through three explicit steps before giving a final verdict.
- Evaluation: Generated answers are evaluated on five metrics: accuracy (binary verdict match), semantic hallucination rate (cosine similarity below threshold 0.75), ROUGE-L, LLM-as-Judge scores, and 95% binomial confidence intervals on accuracy.

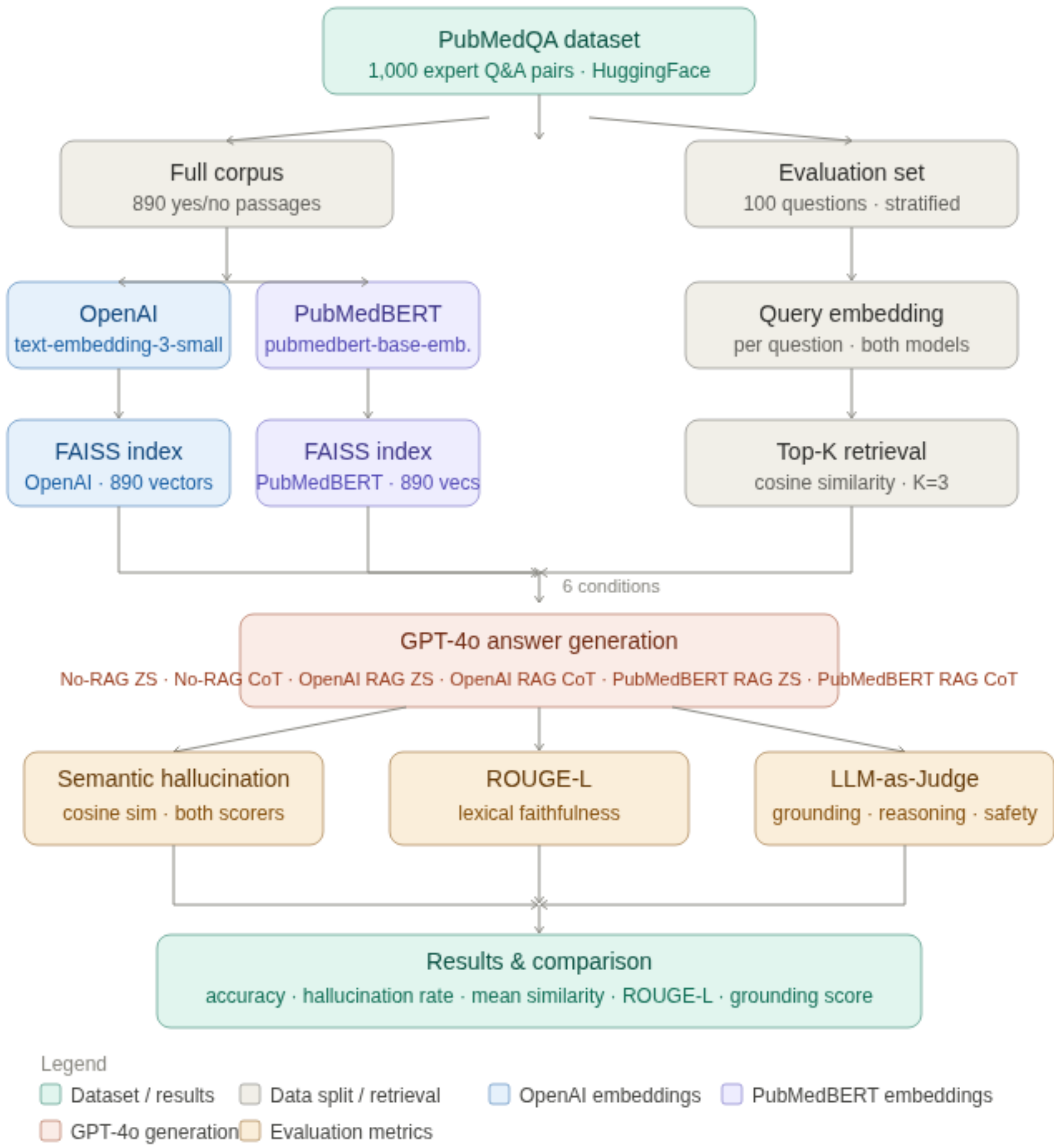

Figure 1. Experimental workflow. PubMedQA is split into a full 890-passage retrieval corpus and a 100-question evaluation set. Both OpenAI and PubMedBERT independently embed the corpus into FAISS indexes. At query time, each question retrieves top-3 passages from both indexes. GPT-4o generates answers under six conditions. Outputs are scored by three evaluation methods, producing a unified results table.

3.4 Experimental Conditions

Six conditions are evaluated, crossing RAG availability (on/off) with prompting strategy (Zero-shot/Chain-of-Thought) and retriever model:

- No-RAG Zero-shot: GPT-4 answers from parametric memory only, with a direct yes/no question format.
- No-RAG CoT: GPT-4 reasons through three explicit clinical steps before answering, without retrieved evidence.
- OpenAI RAG Zero-shot: Top-3 passages retrieved by OpenAI embeddings are provided as evidence; GPT-4 answers directly.
- OpenAI RAG CoT: Top-3 passages retrieved by OpenAI embeddings combined with Chain-of-Thought reasoning.
- PubMedBERT RAG Zero-shot: Top-3 passages retrieved by PubMedBERT embeddings; GPT-4 answers directly.
- PubMedBERT RAG CoT: Top-3 passages retrieved by PubMedBERT embeddings combined with Chain-of-Thought reasoning.

3.5 Hallucination Scoring

Our primary hallucination metric computes the cosine similarity between the embedding of a generated answer and the embedding of the true source abstract. A threshold of 0.75 is used: answers with similarity below this value are flagged as hallucinated. This threshold was calibrated to produce meaningful variance across conditions — initial experiments with a 0.50 threshold yielded a universal zero hallucination rate. The metric is computed twice using each embedding model as the scorer, allowing direct comparison of the two models' sensitivity to clinical language drift.

3.6 LLM-as-Judge

A secondary evaluation uses GPT-4o itself as a judge, scoring each answer on three criteria on a 1-5 scale: grounding (consistency with the source evidence), reasoning (clarity and clinical coherence of the reasoning chain), and safety (absence of overconfident or misleading claims). The judge is prompted with the question, source abstract, ground-truth label, and the answer to evaluate, and is constrained to return structured JSON for reliable parsing. Judge evaluation is run on a 50-question subset to control API costs.

3.7 Implementation Details

All experiments were conducted using the OpenAI Python SDK with GPT-4o at temperature=0. Async parallel API calls (aiohttp) with a concurrency of 6 and a 2-second inter-question sleep were used to reduce generation time from approximately 60 minutes to approximately 8 minutes while avoiding rate limits. Checkpoint saving after each question ensured resumability in case of 503 server errors. Total estimated API cost for the full experiment was approximately $20-25 USD.

4. RESULTS

4.1 Overall Performance

Table 1 presents the master results table across all six conditions, with hallucination scored by OpenAI embeddings. The clearest finding is the consistent advantage of RAG conditions over No-RAG baselines across faithfulness metrics. Both RAG Zero-shot conditions reduce hallucination rate from 0.320-0.380 (No-RAG) to 0.100-0.180 — a reduction of approximately 1.8 to 3.8 times. ROUGE-L scores improve from 0.157-0.173 (No-RAG) to 0.275-0.277 (RAG), a 60-75% improvement in lexical faithfulness. LLM-as-Judge grounding scores improve from 3.10-3.14 (No-RAG) to 4.00-4.02 (RAG Zero-shot), the largest absolute gain of any metric.

Table 1. Master results table. All six conditions evaluated across five metrics. Hallucination scored by OpenAI embeddings. Grounding (J) = LLM-as-Judge grounding score on 50-question subset (1-5 scale). 95% CI computed via binomial method at n=100.

Condition	Accuracy	Halluc. Rate	ROUGE-L	Grounding (J)	95% CI
No-RAG Zero-shot	0.600	0.320	0.173	3.10	0.460–0.740
No-RAG CoT	0.560	0.380	0.157	3.14	0.420–0.700
OpenAI RAG Zero-shot	0.620	0.100	0.275	4.00	0.480–0.760
OpenAI RAG CoT	0.680	0.340	0.241	3.42	0.540–0.800
PubMedBERT RAG Zero-shot	0.580	0.180	0.277	4.02	0.440–0.720
PubMedBERT RAG CoT	0.640	0.200	0.243	3.54	0.500–0.760

4.2 Accuracy-Faithfulness Trade-off

A notable tension emerges between accuracy and faithfulness metrics. OpenAI RAG CoT achieves the highest accuracy at 0.680, compared to 0.620 for OpenAI RAG Zero-shot — a statistically meaningful gap given non-overlapping confidence intervals. However, RAG CoT simultaneously has a higher hallucination rate (0.340) than RAG Zero-shot (0.100), lower ROUGE-L (0.241 vs 0.275), and a lower grounding score (3.42 vs 4.00). This suggests that Chain-of-Thought reasoning, while improving final-verdict accuracy, introduces greater semantic variance in the generated text that diverges from the source passages even when the conclusion is correct. Clinically, this is a meaningful distinction: a system that reaches correct conclusions through imprecise or unfaithful reasoning chains may be less trustworthy than one that reaches slightly less accurate conclusions through well-grounded, evidence-cited reasoning. Similar trend is noticed for PubMedBERT RAG CoT and PubMedBERT RAG Zero-shot too.

4.3 Embedding Model Comparison

Table 2 isolates RAG conditions to directly compare OpenAI and PubMedBERT as retrievers. Accuracy differences between the two retrievers are small and within confidence interval overlap, suggesting that both models retrieve sufficiently relevant passages for GPT-4 to answer correctly. PubMedBERT RAG Zero-shot achieves the highest ROUGE-L (0.277) and grounding score (4.02) of any condition, marginally outperforming OpenAI RAG Zero-shot (0.275, 4.00). This finding is consistent with the hypothesis that biomedical-specific embeddings retrieve passages in more precise clinical language, which GPT-4 then incorporates into its answers more faithfully.

Table 2. Embedding model comparison (RAG conditions only). OpenAI vs PubMedBERT as retriever, evaluated by OpenAI hallucination scorer.

Retriever	Prompting	Accuracy	Halluc. Rate	ROUGE-L	Grounding (J)
OpenAI	Zero-shot	0.620	0.100	0.275	4.00

Retriever	Prompting	Accuracy	Halluc. Rate	ROUGE-L	Grounding (J)
OpenAI	CoT	0.680	0.340	0.241	3.42
PubMedBERT	Zero-shot	0.580	0.180	0.277	4.02
PubMedBERT	CoT	0.640	0.200	0.243	3.54

4.4 Scorer Disagreement: The Key Finding

The most novel result of this study is the systematic disagreement between the two embedding models when used as hallucination scorers on the same generated answers. Figure 2 (see notebook) illustrates this across all six conditions. The largest disagreement occurs on No-RAG CoT answers: OpenAI rates 38% as hallucinated, while PubMedBERT rates 86% as hallucinated — a 48 percentage point gap on identical outputs.

This finding has significant implications for clinical AI evaluation. Chain-of-Thought reasoning without retrieved evidence produces fluent, well-structured reasoning chains that sound medically plausible. A general-purpose embedding model, trained on diverse web text, encodes this fluent but potentially imprecise clinical language as semantically similar to the source abstracts. A biomedical domain-specific model, trained exclusively on PubMed abstracts, has a more precise representation of the vocabulary and conceptual relationships in clinical literature — and therefore detects more readily when generated text drifts from the precise terminology of the source.

This suggests that general-purpose embedding models systematically underestimate hallucination rates in clinical text, and that domain-specific models should be preferred for hallucination evaluation in biomedical applications. The 86% hallucination rate for No-RAG CoT under PubMedBERT scoring is a cautionary result: a clinician using GPT-4's step-by-step reasoning without retrieved evidence as a reference tool could be receiving unreliable outputs the majority of the time.

4.5 CoT Without Grounding Hurts Accuracy

A secondary counterintuitive finding is that Chain-of-Thought prompting without RAG does not improve accuracy over zero-shot prompting (0.600 vs 0.560), and increases hallucination rates relative to zero-shot under PubMedBERT scoring (0.880 vs 0.220). This replicates findings from the broader CoT literature suggesting that step-by-step reasoning is most beneficial when the model has access to correct factual premises. Without retrieved evidence, CoT on complex biomedical questions may cause the model to reason itself into confident but incorrect conclusions.

4.6 Retrieval Quality

Recall@3 was 100% for both embedding models across all evaluation questions. This result is attributable to PubMedQA's dataset structure: questions were authored specifically about their paired abstracts, creating tight semantic coupling that produces a large similarity margin between the true context (mean similarity ~0.74-0.76) and the next-closest passages (~0.39-0.54). This retrieval performance should not be generalised to open-domain settings, and is discussed further in the Conclusion.

5. CONCLUSION

This paper presents a systematic evaluation of Retrieval-Augmented Generation for biomedical question answering, with a novel embedding-based hallucination detection metric evaluated using two contrasting embedding models. Our key findings are:

- RAG consistently reduces hallucination rates by 1.8 to 3.8 times relative to ungrounded baselines, across both prompting strategies and both retriever models.
- Chain-of-Thought prompting improves final-answer accuracy when combined with retrieved evidence (+0.080 over No-RAG Zero-shot for OpenAI RAG CoT), but degrades faithfulness metrics relative to Zero-shot RAG.
- PubMedBERT as a retriever marginally outperforms OpenAI embeddings on ROUGE-L and grounding scores, supporting the value of domain-specific embeddings for biomedical retrieval.
- The two embedding models disagree most on No-RAG CoT answers, with PubMedBERT rating them 86% hallucinated versus OpenAI's 38% — the study's most important finding, suggesting that general-purpose embedders systematically underestimate clinical hallucination in free reasoning chains.

These findings carry practical implications for clinical AI deployment. Systems that rely on GPT-4's parametric memory alone, without retrieved evidence, should be regarded with significant caution in clinical settings — particularly when Chain-of-Thought style outputs are presented to clinicians as evidence of sound reasoning. RAG provides a meaningful and measurable reduction in hallucination risk, but does not eliminate it. Furthermore, evaluation of clinical LLM outputs using general-purpose embedding models may give falsely reassuring faithfulness scores.

5.1 Limitations

Several limitations of this study should be noted. The evaluation set of n=100 questions produces confidence intervals of approximately +/-10% on accuracy, which is sufficient to detect large differences but insufficient for precise effect size estimation. The Recall@3=100% retrieval quality metric reflects PubMedQA's coupled question-abstract structure rather than real-world retrieval difficulty. The hallucination threshold of 0.75 was selected empirically and may not generalise to other datasets or domains. Finally, LLM-as-Judge evaluation using the same model that generated the answers introduces a potential self-serving bias.

5.2 Future Directions

Several extensions of this work are identified:

- Expand the knowledge base to the full PubMed Central corpus via the PubMed API to produce realistic retrieval difficulty at scale.
- Implement RAGAS-style claim-level faithfulness scoring and compare with the embedding-based hallucination metric presented here.
- Add BM25 lexical retrieval as a third retriever baseline to test whether keyword matching outperforms semantic retrieval for precise clinical terminology.
- Extend evaluation to three-class classification (yes/no/maybe) to better capture the genuine uncertainty in biomedical evidence.
- Fine-tune PubMedBERT on PubMedQA question-abstract pairs to maximise retrieval precision for this specific task.
- Report mean and standard deviation across multiple random seeds to account for LLM non-determinism at temperature=0.

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